01. Business Problem & Objective

Define the business decision, then map it to an unsupervised learning problem.

Project: SmartCart Customer Segmentation

Source checked: Sai-Likhith-2005/Customer-Segmentation | smartcart.ipynb | smartcart_customers.csv

Business context

The supplied project is a retail customer-segmentation exercise using customer demographics, household information, spending, purchase channels and campaign-response fields. The goal is to group customers with similar observed profiles so the business can understand its customer base and design differentiated marketing experiments.

ML formulation

Element	Definition
Learning type	Unsupervised learning
Observation	One customer record
Inputs used in notebook	15 cleaned fields, expanded to 18 columns after one-hot encoding
Algorithmic representation	Standardized feature matrix, then 3-dimensional PCA representation
Clustering	K-Means for choosing K; Agglomerative Clustering for final labels
Output	A cluster label for each of 2,236 retained records

What segmentation can and cannot do

- It can identify groups that are similar under the chosen features and distance metric.
- It can support hypotheses about customer needs, channel preferences and spending behaviour.
- It does not discover objectively true customer types; results depend on feature selection, scaling, PCA, distance and K.
- It does not prove that a marketing action will cause higher sales. That requires controlled experiments.

Business questions worth asking

- Which groups spend more or less, and on which product categories?
- Which groups prefer web, catalogue or store purchases?
- Do groups differ in deal-seeking, recency or campaign response?
- Can a group be targeted with a relevant offer without wasting budget or over-contacting customers?

Critical framing

The repository clusters all available records, then describes cluster averages. Treat labels such as “high-spend” or “low-spend” as descriptive hypotheses, not fixed customer identities or proof of future behaviour.